2542_DIT_ITSD001_Main EQP v1.pdf（22 October 2025）

Question 1

(a)

Mr Tan is opening an online store specializing in refurbished laptops. To understand the market price range, he visited 10 different online marketplaces and collected the prices (in $'00) of laptops with similar specifications:

12	**18**	**18**	**23**	**29**	**35**	**40**	**42**	**47**	**55**

Compute the mean, median, mode, range and standard deviation. (Express your answer to 2 decimal places).
(10 marks)

1. **Answer: Mean = 31.90, Median = 32.00, Mode = 18.00, Range = 43.00, Sample Standard Deviation = 14.18.**

   考察知识点：第4章 Numerical Descriptive Measure。

   解题步骤：

   1. 先把数据按从小到大排列：
      12, 18, 18, 23, 29, 35, 40, 42, 47, 55
   2. 求平均数：
      $$\bar{x} = \frac{12+18+18+23+29+35+40+42+47+55}{10} = \frac{319}{10} = 31.9$$
   3. 求中位数：
      共有 10 个数据，取第5个和第6个数据的平均值。
      第5个是 29，第6个是 35。
      $$\text{Median} = \frac{29+35}{2} = 32$$
   4. 求众数：
      出现次数最多的是 18，所以众数是 18。
   5. 求极差：
      $$\text{Range} = 55 - 12 = 43$$
   6. 求样本标准差：
      这里把 10 个价格当作样本来处理，所以用样本标准差公式。

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

代入计算后：

$$s = \sqrt{\frac{1808.90}{9}} = 14.18$$

(b)

Sharon wants to estimate the monthly household expenses on electricity. She conducted a small survey with 10 families in her neighborhood. Their monthly bills (in $) are as follows:

Amount Spent ($)	85	92	95	105	108	111	117	118	134	138

Construct a stem and leaf diagram from the above data. Use stem unit = $10 and leaf unit =$**1.**
(6 marks)

茎叶图绘制 (Stem-and-Leaf Diagram)

考察知识点： Topic 3: Organising and Visualising Variables

解题思路：

茎叶图的目的是展示数据分布。题目要求“茎单位为10”，这意味着我们要把数字拆开：十位及以上作为“茎”（Stem），个位作为“叶”（Leaf）。

解题步骤：

- **第一步：列出茎。** 数据范围从 85 到 138，所以茎应该是 8, 9, 10, 11, 12, 13。
- **第二步：填入叶子。** 比如 85，茎是 8，叶是 5。注意：即便 120-129 之间没有数据，茎“12”也必须写出来，空出叶子部分，以体现分布的断层。
- **第三步：添加图例 (Key)。** 这是一个必得分点，告诉阅卷老师 8 | 5 代表什么。

Answer:

Stem-and-Leaf Display (Stem unit = 10, Leaf unit = 1)

Stem	Leaf
8	5
9	2 5
10	5 8
11	1 7 8

Stem	Leaf
12	
13	4 8

(c)

Mr Tan also sells tablets in his store. He recorded the number of tablets sold daily over a 20-day period:

Number of tablets sold	Number of days (Frequency)	Relative Frequency (%)	Class Mid-point
0 up to 3	4		
3 up to 6	7		
6 up to 9	5		
9 up to 12	3		
12 up to 15	1		
Total	20		

Based on the frequency distribution table above, calculate the relative frequencies (Express your answer to 2 decimal places) and class mid-points. Construct a histogram based on the frequency distribution table provided. Label all axes clearly.
(9 marks)

Answer:

Class Interval	Frequency	Relative Frequency (%)	Class Mid-point
0 up to 3	4	20.00%	1.5
3 up to 6	7	35.00%	4.5
6 up to 9	5	25.00%	7.5
9 up to 12	3	15.00%	10.5
12 up to 15	1	5.00%	13.5
Total	20	100.00%	

Histogram sketch:

```
Frequency
7 |       ▇
6 |       ▇
5 |     ▇ ▇
4 | ▇   ▇ ▇
3 | ▇   ▇ ▇ ▇
2 | ▇   ▇ ▇ ▇
1 | ▇   ▇ ▇ ▇ ▇
  +-------------------------
    0-3  3-6  6-9  9-12 12-15
      Number of tablets sold
```

考察知识点：第3章 Organising and Visualising Variables。

解题步骤：

1. relative frequency 的公式是：

$$\text{Relative Frequency} = \frac{\text{Frequency}}{20} \times 100\%$$

2. 逐组计算：
   - 4/20 = 20.00%
   - 7/20 = 35.00%
   - 5/20 = 25.00%
   - 3/20 = 15.00%
   - 1/20 = 5.00%
3. class mid-point 的公式是：

$$\text{Mid-point} = \frac{\text{lower limit} + \text{upper limit}}{2}$$

4. 逐组计算：
   - $(0+3)/2 = 1.5$
   - $(3+6)/2 = 4.5$
   - $(6+9)/2 = 7.5$
   - $(9+12)/2 = 10.5$
   - $(12+15)/2 = 13.5$
5. 画直方图时，每一组的柱高就是频数，组距相同，所以可以直接按 4, 7, 5, 3, 1 画柱子。

Question 2

(a)

A technology company is hiring 10 data analyst interns from a pool of 200 applicants. Applicants are classified by education level and preferred department as shown below.

Department	Diploma Holders	Degree Holders	Total
IT	45	35	80
Marketing	25	40	65
Finance	15	40	55
Total	85	115	200

Given that the applicant is from marketing, what is the probability that the applicant is a degree holder? What is the probability that an applicant is from IT or the applicant is a diploma holder? (Express answers to 3 decimal places)
(8 marks)

Answer: P(Degree Holder | Marketing) = 0.615. P(IT or Diploma Holder) = 0.600.

考察知识点：第5章 Basic Probability。

解题步骤：

1. 先求条件概率：

$$P(\text{Degree}|\text{Marketing}) = \frac{P(\text{Degree and Marketing})}{P(\text{Marketing})}$$

2. 从表中读数：
   - Marketing 且 Degree = 40
   - Marketing 总数 = 65
3. 代入：

$$P(\text{Degree}|\text{Marketing}) = \frac{40}{65} = 0.615$$

4. 再求 “IT or Diploma Holder”：
   用加法公式：

$$P(IT \text{ or } Diploma) = P(IT) + P(Diploma) - P(IT \text{ and } Diploma)$$

5. 代入表中数据：

$$P(IT) = \frac{80}{200}, \quad P(Diploma) = \frac{85}{200}, \quad P(IT \text{ and } Diploma) = \frac{45}{200}$$

6. 所以：

$$P(IT \text{ or } Diploma) = \frac{80 + 85 - 45}{200} = \frac{120}{200} = 0.600$$

(b)

The technology company also wants to estimate monthly server maintenance costs. Historical data suggest the cost follows a normal distribution with mean $\mu = 1200$ and standard deviation $\sigma = 150$, what is $P(1200 < X < 1300)$? If the company randomly samples 25 months, calculate the probability that the average monthly cost exceeds $1,230? (Express answer to 3 decimal places)
(8 marks)

Answer: P(1200 < X < 1300) = 0.248. P(\bar{X} > 1230) = 0.159.

考察知识点：第7章 Continuous Distributions and Normal Distributions；第8章 Sampling Distributions and Central Limit Theorem。

解题步骤：

1. 已知：

$$\mu = 1200, \quad \sigma = 150$$

2. 先算 $P(1200：
   把 X 标准化：

$$z = \frac{x - \mu}{\sigma}$$

3. 当 $X = 1200$ 时：

$$z = \frac{1200 - 1200}{150} = 0$$

4. 当 $X = 1300$ 时：

$$z = \frac{1300 - 1200}{150} = 0.6667$$

5. 所以：

$$P(1200 < X < 1300) = P(0 < Z < 0.6667) = 0.2475 \approx 0.248$$

6. 再算样本均值超过 1230 的概率：
   样本量 $n = 25$
7. 标准误差：

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{150}{\sqrt{25}} = 30$$

8. 计算 z 值：

$$z = \frac{1230 - 1200}{30} = 1$$

9. 所以：

$$P(\bar{X} > 1230) = P(Z > 1) = 1 - 0.8413 = 0.1587 \approx 0.159$$

(c)

The technology company organized a team-building event featuring a lucky spin game for the new hires. Each play costs $50. If a star appears $(P = 1/5)$, the player gets $100. If a moon appears $(P = 1/20)$, the player gets $200. If a crown appears $(P = 1/100)$, the player gets $500. Otherwise, the player gets $0. What amount can a player expect to win or lose by playing the game?
(9 marks)

Answer: The expected net result is a loss of $15.00 per play.

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 先算每种结果的期望奖金：

$$E(X) = 100\left(\frac{1}{5}\right) + 200\left(\frac{1}{20}\right) + 500\left(\frac{1}{100}\right) + 0$$

2. 分别计算：

$$100 \cdot \frac{1}{5} = 20$$

$$200 \cdot \frac{1}{20} = 10$$

$$500 \cdot \frac{1}{100} = 5$$

3. 所以期望奖金是：

$$20 + 10 + 5 = 35$$

4. 但每次玩游戏要先付 50 元，所以净结果是：

$$35 - 50 = -15$$

5. 负号表示亏损，所以玩家平均每次会亏 15 元。

Question 3

(a)

A customer service center receives an average of 3 complaints per hour, and the number of complaints follows a Poisson distribution. Find the probability that in a particular hour the center receives:

(i) Exactly 4 complaints.
(ii) At least 2 complaints. (Correct your answers to 4 decimal places)
(10 marks)

Answer:
(i) P(X = 4) = 0.1680
(ii) P(X \ge 2) = 0.8009

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 这是 Poisson distribution。
   已知平均每小时投诉数：

$$\lambda = 3$$

2. Poisson 公式：

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

3. 第(i)问，求恰好 4 次投诉：

$$P(X = 4) = \frac{e^{-3}3^4}{4!}$$

$$= \frac{e^{-3} \cdot 81}{24} = 0.1680$$

4. 第(ii)问，求至少 2 次投诉：
   “at least 2” 表示 $X \geq 2$
5. 用补事件更方便：

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

6. 代入公式：

$$P(X = 0) = e^{-3}$$

$$P(X = 1) = 3e^{-3}$$

7. 所以：

$$P(X \geq 2) = 1 - e^{-3} - 3e^{-3} = 1 - 4e^{-3} = 0.8009$$

(b)

A battery factory produces rechargeable batteries, where 1 in 8 is defective. 16 batteries are tested at random. Find the probability that less than three of them will be defective. (Correct your answer to 4 decimal places)
(9 marks)

Answer: P(X < 3) = 0.6771.

考察知识点：第6章 Discrete Probability Distributions。

解题步骤：

1. 这是 binomial distribution。
   设：

$$n = 16, \quad p = \frac{1}{8}, \quad q = 1 - p = \frac{7}{8}$$

2. “less than three” 表示：

$$X = 0, 1, 2$$

3. 所以要算：

$$P(X < 3) = P(0) + P(1) + P(2)$$

4. 用二项分布公式：

$$P(X = x) = \binom{16}{x}\left(\frac{1}{8}\right)^x\left(\frac{7}{8}\right)^{16-x}$$

5. 逐项计算并相加：

$$P(X < 3) = \binom{16}{0}\left(\frac{1}{8}\right)^0\left(\frac{7}{8}\right)^{16} + \binom{16}{1}\left(\frac{1}{8}\right)^1\left(\frac{7}{8}\right)^{15} + \binom{16}{2}\left(\frac{1}{8}\right)^2\left(\frac{7}{8}\right)^{14}$$

6. 结果为：

$$P(X < 3) = 0.6771$$

(c)

State the level of measurement (Ratio, Ordinal or Nominal) for the variables: “Number of online purchases per month” and “Favourite smartphone brand”.
(6 marks)

Answer: Number of online purchases per month = Ratio. Favourite smartphone brand = Nominal.

考察知识点：第2章 Defining and Collecting Data。

解题步骤：

1. “Number of online purchases per month” 是计数型变量。
   它有真实的 0，而且 10 次比 5 次正好是 2 倍，所以是 ratio scale。
2. “Favourite smartphone brand” 是品牌名称分类。
   只是分类，没有大小顺序，所以是 nominal scale。

Question 4

(a)

A survey of 20 software engineers found their average weekly coding hours to be 45.3, with sample standard deviation 8.2 hours. Find the 99% confident interval of the population mean. (Express your answer to 3 decimal places)
(9 marks)

Answer: The 99% confidence interval is (40.054, 50.546).

考察知识点：第9章 Confidence Interval and Point Estimation。

解题步骤：

1. 已知：

$$\bar{x} = 45.3, \quad s = 8.2, \quad n = 20$$

2. 因为总体标准差未知，所以用 t interval。
3. 自由度：

$$df = n - 1 = 19$$

4. 99% 置信水平对应：

$$t_{0.005,19} = 2.861$$

5. 标准误差：

$$SE = \frac{s}{\sqrt{n}} = \frac{8.2}{\sqrt{20}} = 1.834$$

6. 误差界：

$$ME = t \times SE = 2.861 \times 1.834 = 5.246$$

7. 置信区间：

$$45.3 \pm 5.246$$

$$= (40.054, 50.546)$$

(b)

Another 30 engineers were surveyed, and the mean and standard deviation remained the same. Find the 95% confident interval for the population mean hours. (Express your answer to 3 decimal places.) (9 marks)

Answer: The 95% confidence interval is (42.238, 48.362).

考察知识点：第9章 Confidence Interval and Point Estimation。

解题步骤：

1. 题目说样本量变成 30，但均值和标准差仍然一样：

$$\bar{x} = 45.3, \quad s = 8.2, \quad n = 30$$

2. 自由度：

$$df = 29$$

3. 95% 置信水平对应：

$$t_{0.025,29} = 2.045$$

4. 标准误差：

$$SE = \frac{8.2}{\sqrt{30}} = 1.497$$

5. 误差界：

$$ME = 2.045 \times 1.497 = 3.062$$

6. 置信区间：

$$45.3 \pm 3.062$$

$$= (42.238, 48.362)$$

(c)

The surveyor has made a mistake in calculating the standard deviation and the correct sample standard deviation has a smaller value of 5. How would this affect the width of the confidence interval? (7 marks)

答案 (Answer):

The width of the confidence interval will decrease.

考察知识点:

Topic 9 Overview: Confidence Interval and Point Estimation (标准差的影响)

解题步骤:

1. 置信区间的宽度由公式 $2 \times t \times \frac{S}{\sqrt{n}}$ 决定。
2. 从公式可以看出，宽度与样本标准差 S 成正比。
3. 当 S 从 8.2 变小为 5 时，分子变小，导致整个边际误差减小。
4. 因此，置信区间会变窄 (Narrower)。

END OF PAPER
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